SLIIT UNI

THE KNOWLEDGE UNIVERSITY

IE 2004 Computer Networks

01 - Introduction

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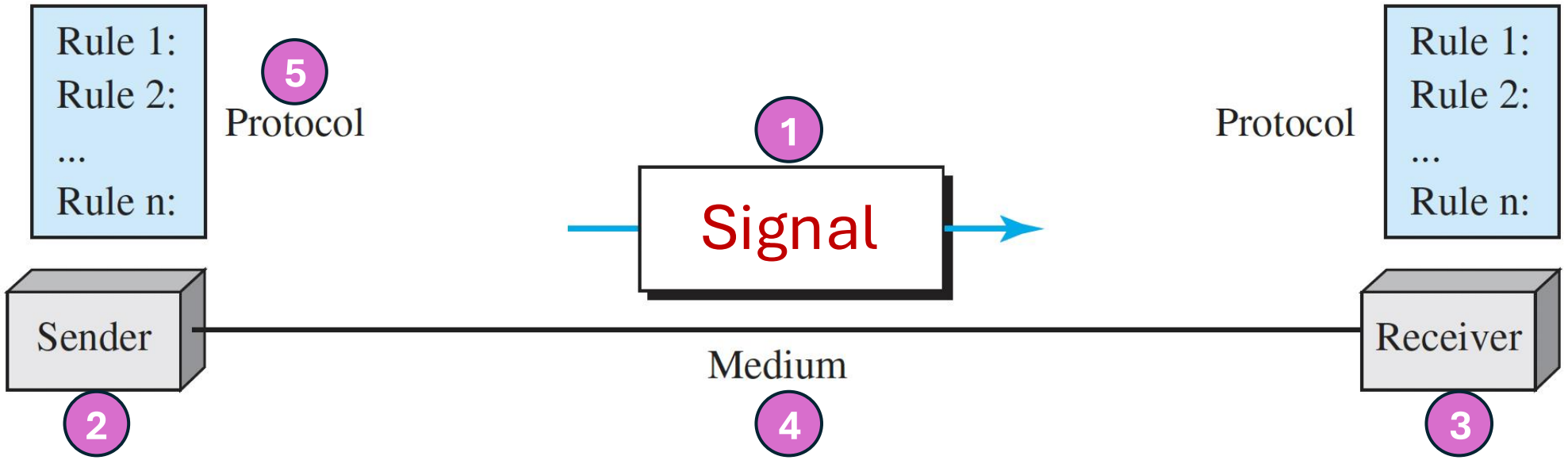
2026 July

Lesson outline

- What is a network?
- Network extent (sizes)
- Network topologies
- Data flow in networks
- Circuit switching and packet switching
- Performance: Loss, Delay, Throughput

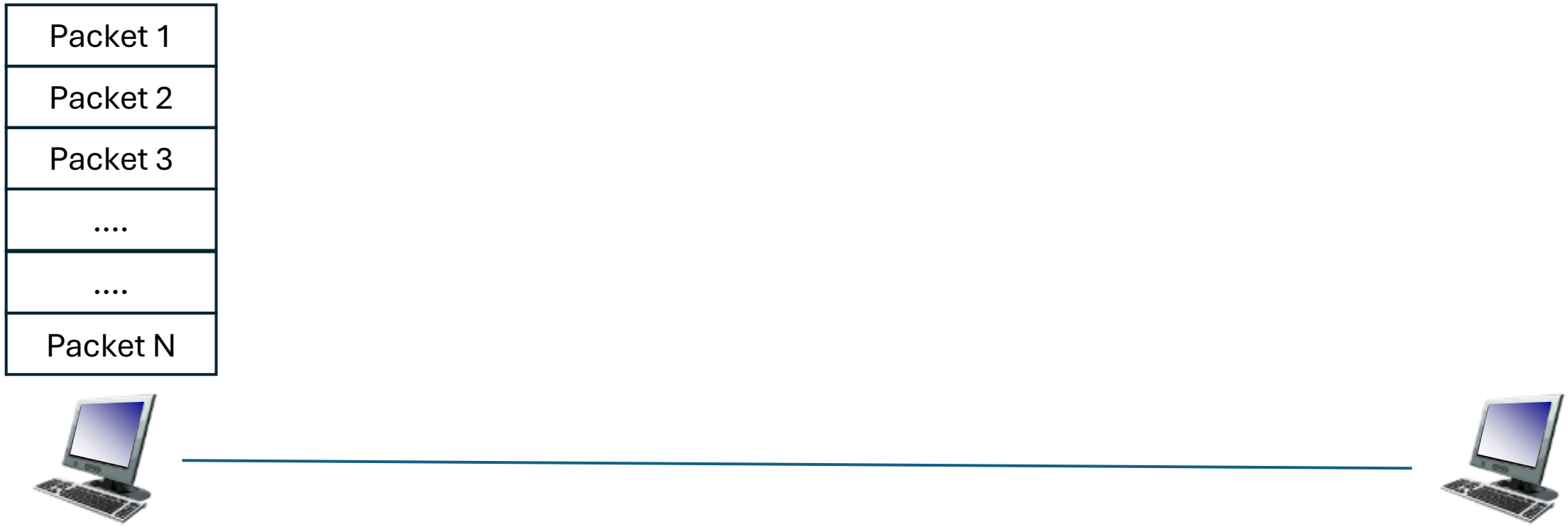
Communication model

- Message: What is communicated (text, numbers, image, video, …)
- Data: Message represented using 1s and 0s
- Signals: Electrical or electromagnetic representations of data
- The signal (representation of the data) travels over the transmission medium (copper wire, optical fiber, free space, …)



Data transmitted in packets

- Message size can vary: few bytes to hundreds of gigabytes or more
- In data communication, data is transmitted in small packets
  - Large messages will be transmitted as multiple packets



Advantages of sending data in packets

- Efficiency: Easier to handle small packets of data
- Error control: If one packet is lost/damaged, only that packet can be resent
- Faster: Different packets can go via different paths



What is a network?

- A network is an interconnection of a set of devices capable of communication
  - Examples: A large computer, desktop, laptop, workstation, mobile devices, a security system, …
  - Internet of Things (IoT)
- Use the term host or end-system to refer to an endpoint device
- There are other devices that perform additional functions
  - Switch: interconnects endpoints to form a small network
  - Router: interconnects networks to other networks
- Node is a common term used to refer to any network device
  - Host, Switch, Router or any other device

“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt: monitor energy use

bikes





cars



scooters



Security Camera



Slingbox: remote control cable TV



Web-enabled toaster + weather forecaster



AR devices



Fitbit



Internet phones



Gaming devices



sensorized bed mattress



diapers

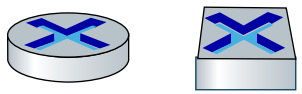
Others?

The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet’s “edge”



Packet switches: forward packets (chunks of data)

- *routers*, *switches*

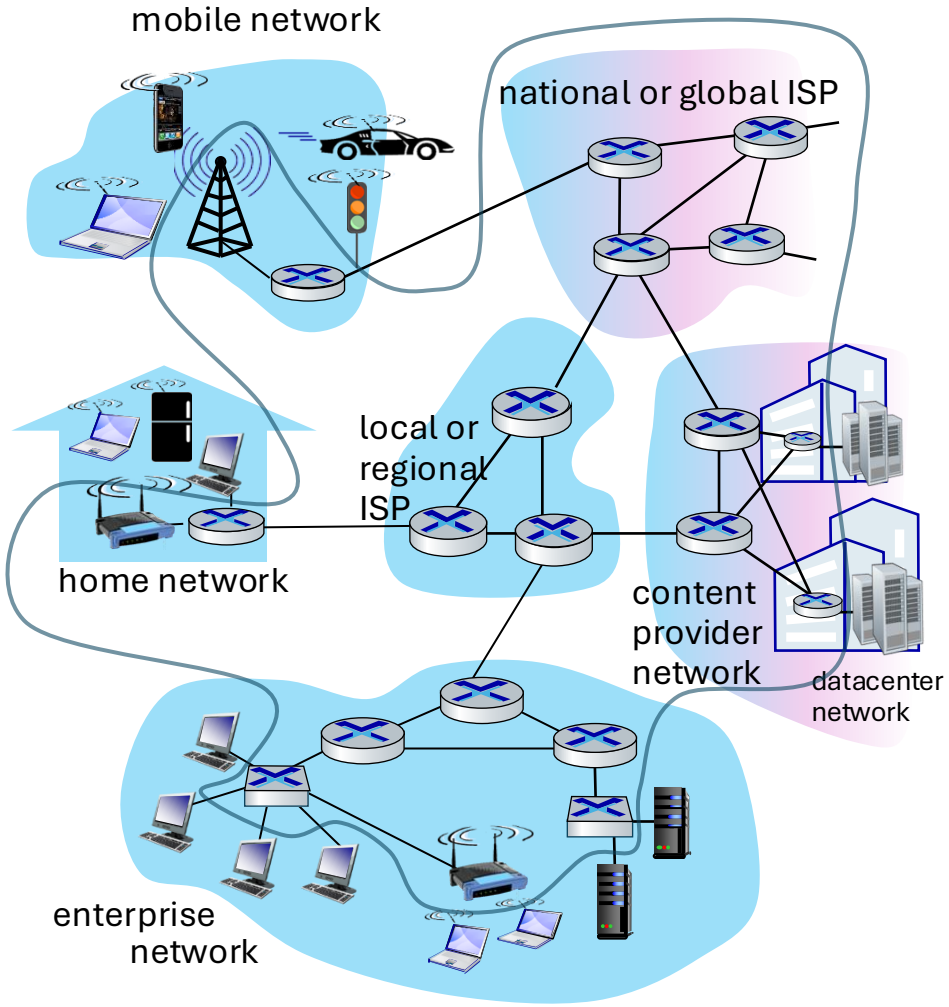


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth* (expressed in bps, kbps, Mbps)

Networks

- collection of devices, routers, links: managed by an organization



Lesson outline

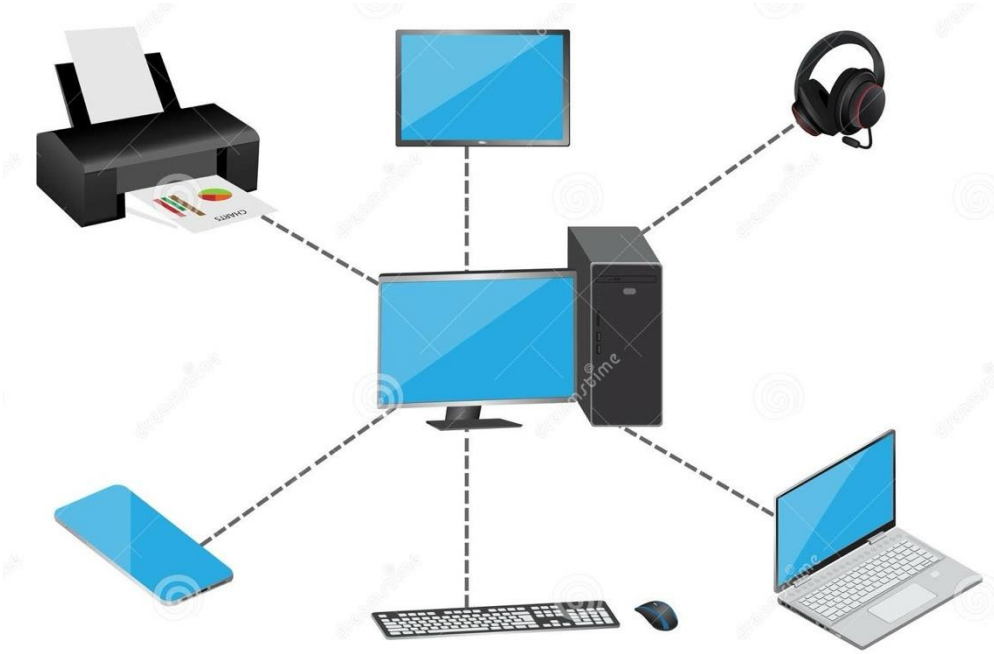
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Network extent

- Personal area network (PAN)
- Local area network (LAN)
- Campus network
- Metropolitan area network (MAN)
- Wide area network (WAN)
- Internet

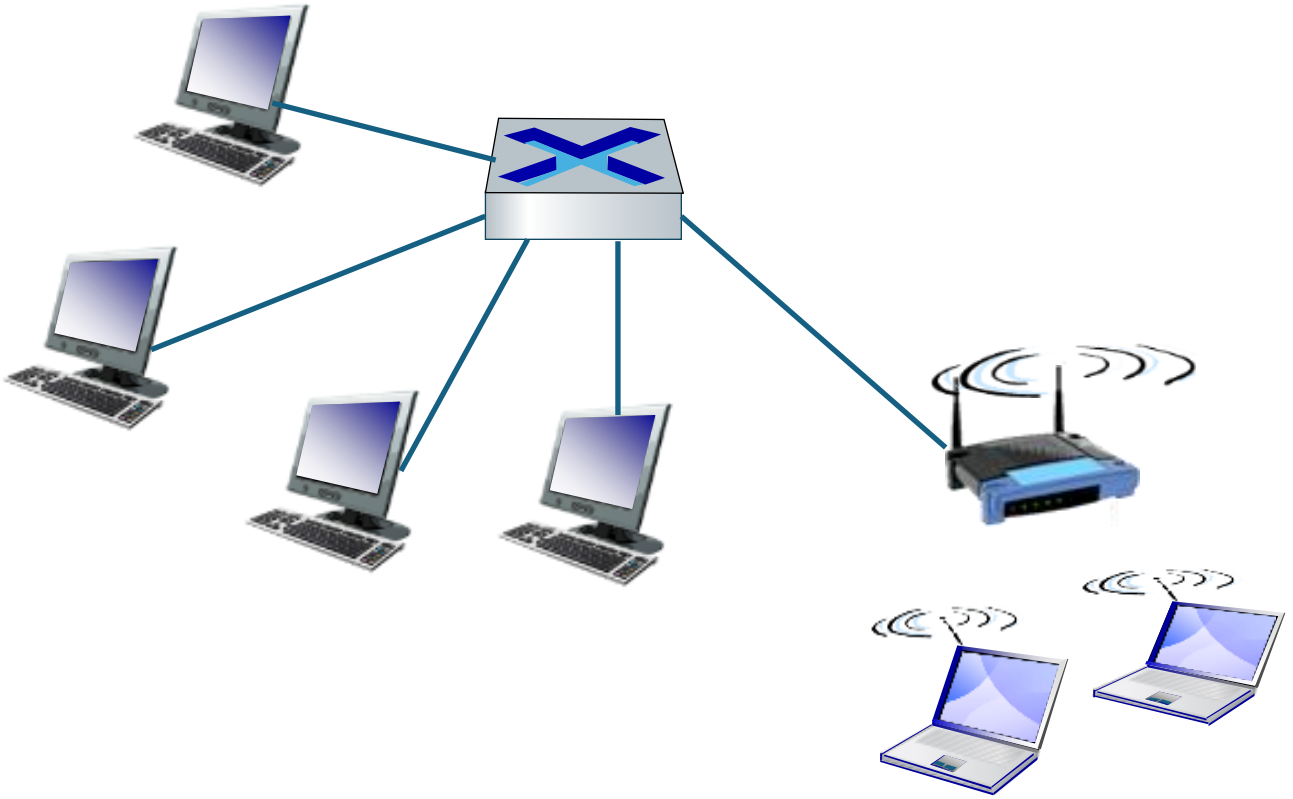
Personal area network (PAN)

- Network connections between devices used by a single person or a small group of persons
- Bluetooth / IrDA
  - Phone connected to a headset
  - Mouse connected to a computer
  - Phone connected to smart watch
  - …
- Network diameter: 1 ~ 5 meters



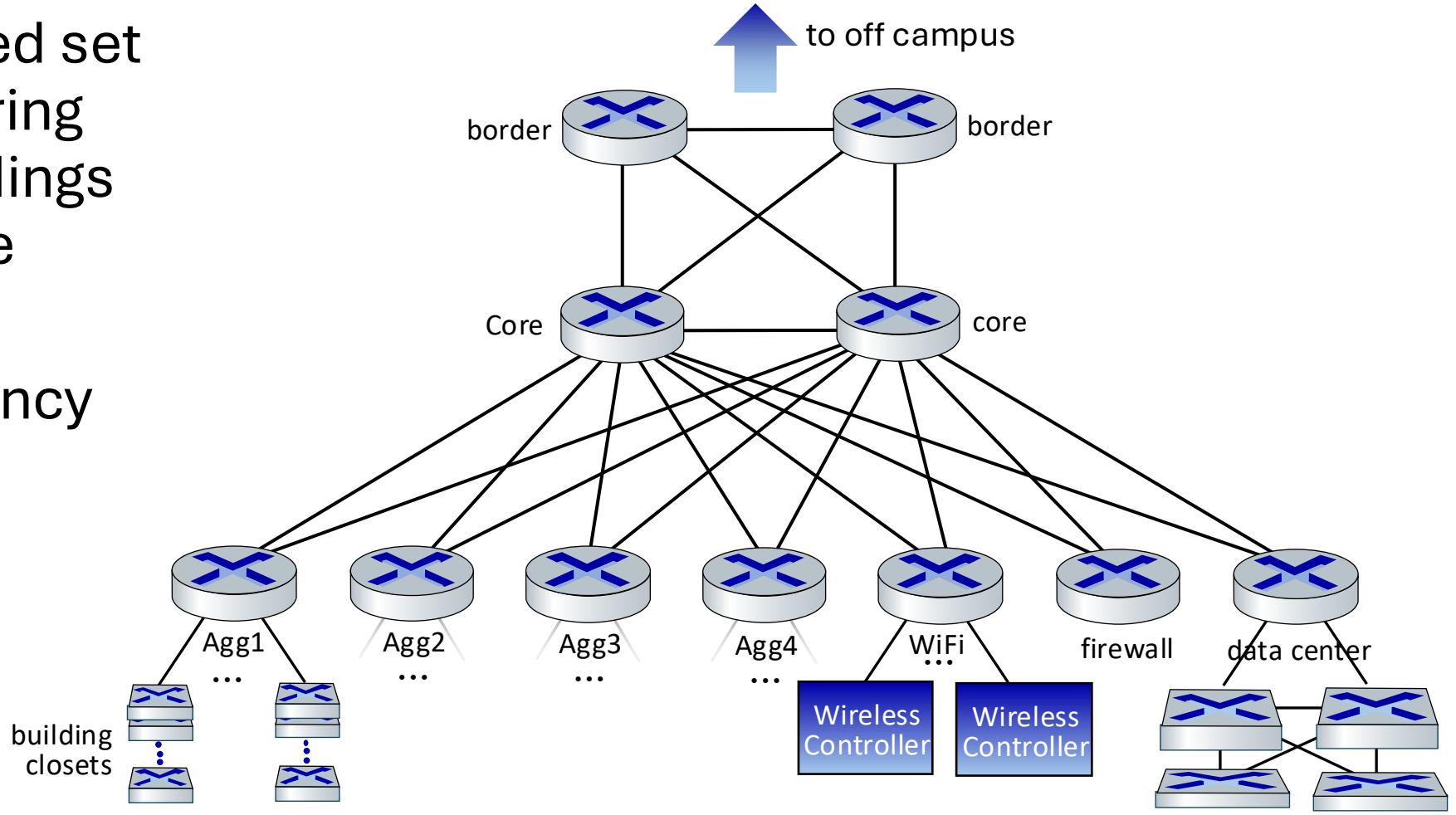
Local area network (LAN)

- Wired and/or wireless
- Covers a home, small office, single floor or part of a floor in a building
- Diameter:
  10m - 200m



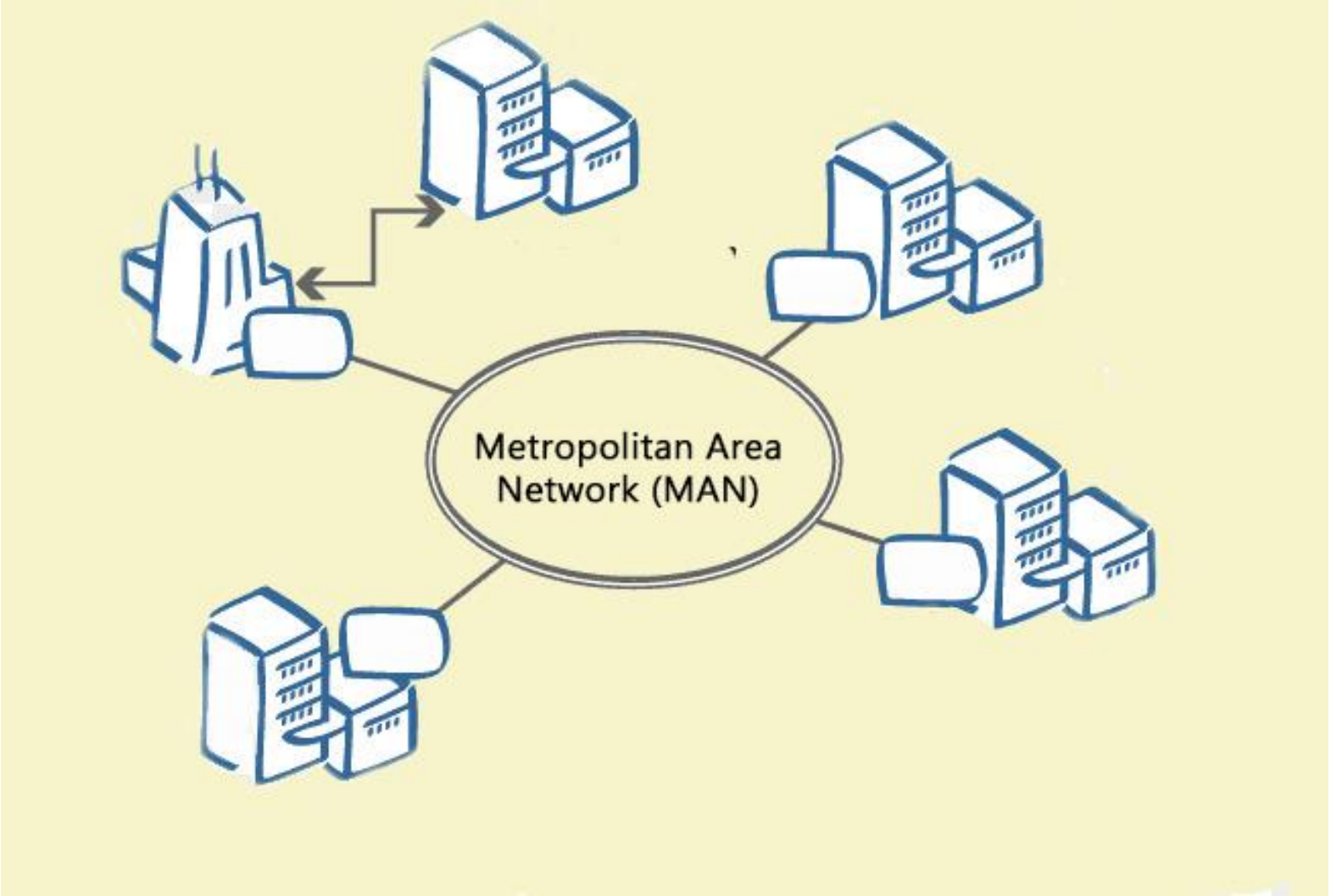
Campus network

- Interconnected set of LANs covering multiple buildings of a single site
- Hierarchical, Has Redundancy
- Diameter: 1 ~ 10 km



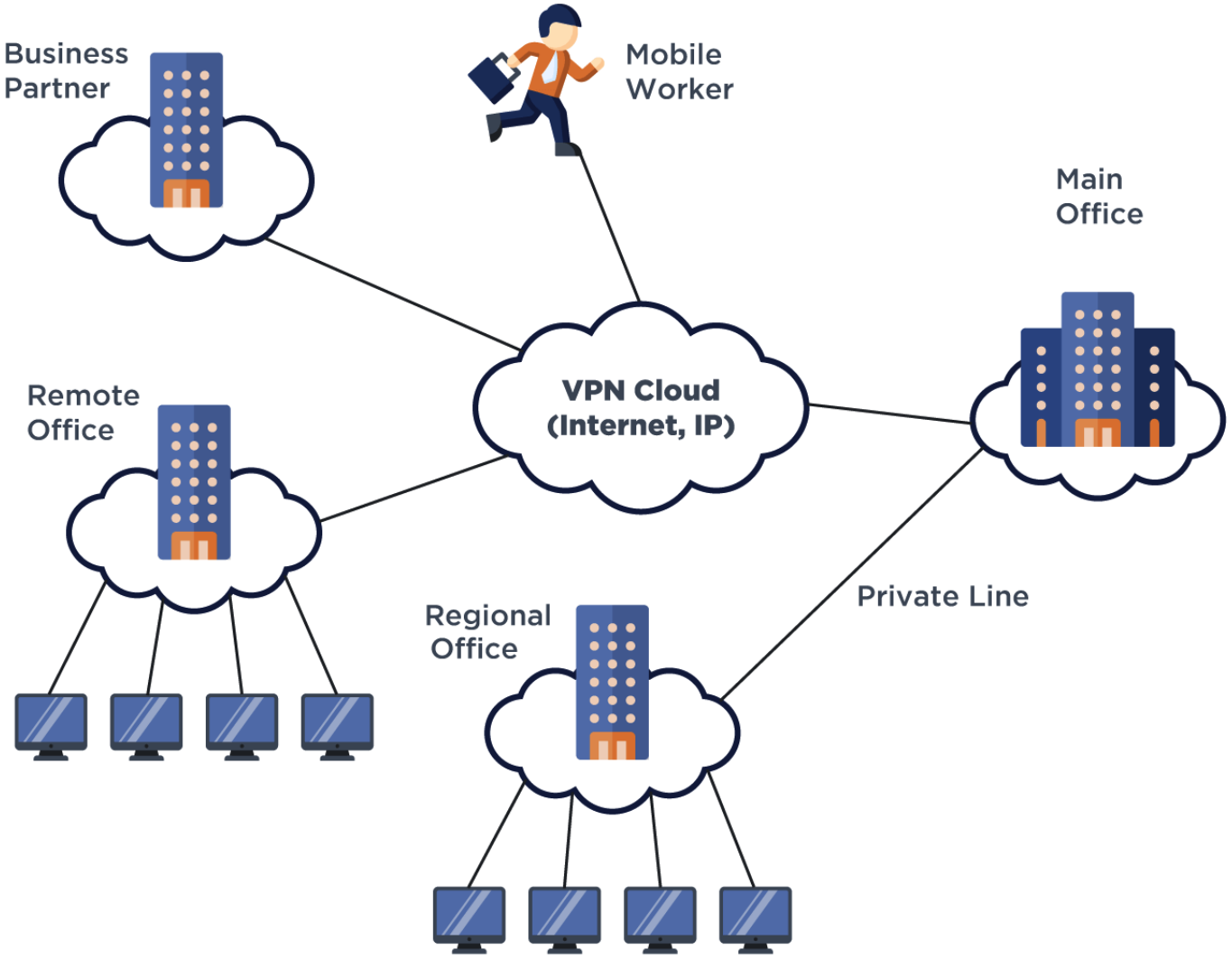
Metropolitan area network (MAN)

- Covers a single metropolitan area (city)
- Usually owned by a service provider
- Diameter: 10 ~ 50 km



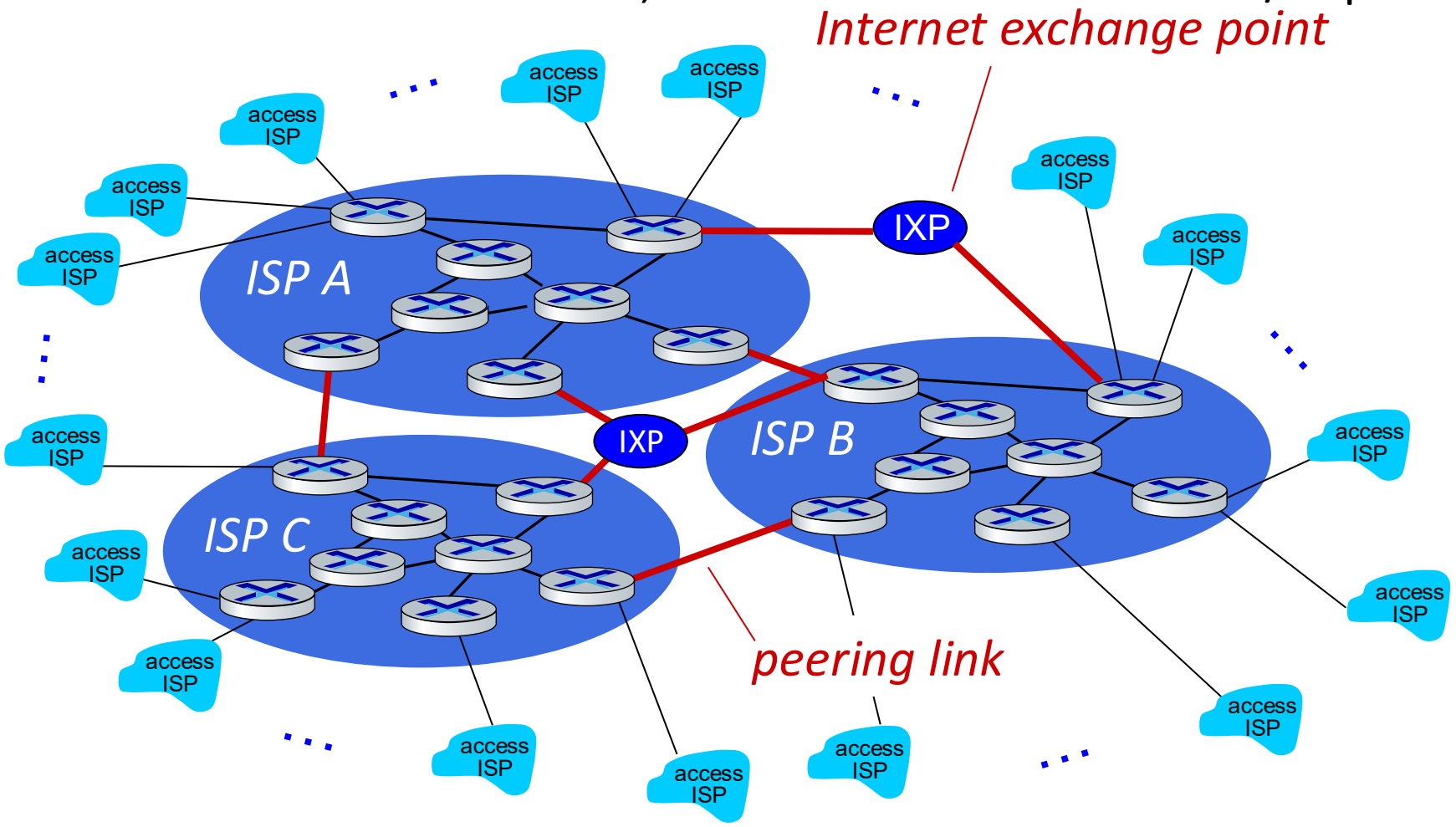
Wide area network (WAN)

- Usually built using point-to-point connections between multiple sites
- Diameter:
  10 km ~ 1000km or more



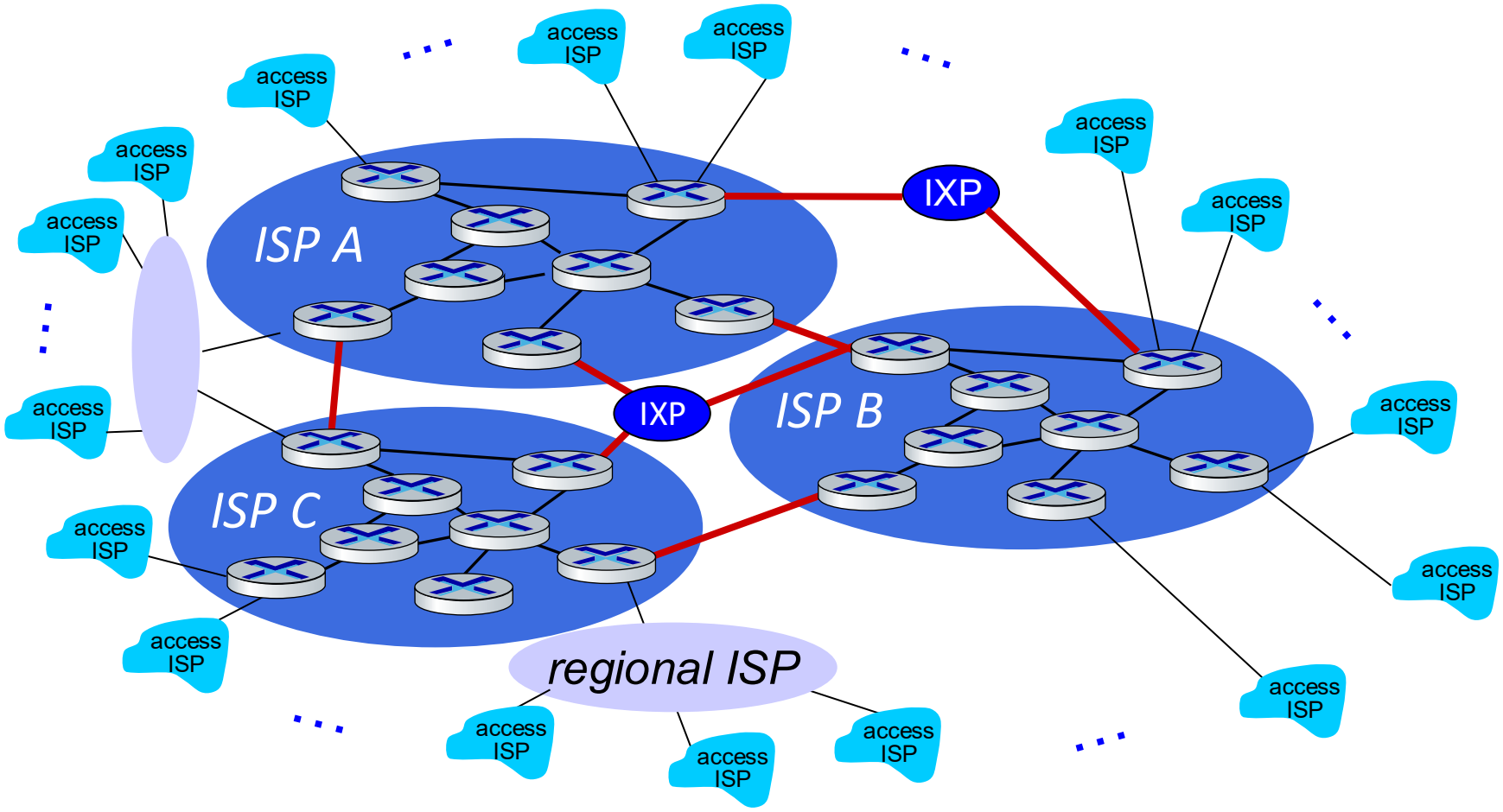
Internet structure: a “network of networks”

ISPs providing access to customers get Internet service from larger ISPs called Tier 1 ISPs or Global ISPs, interconnected via IXPs and/or peering links



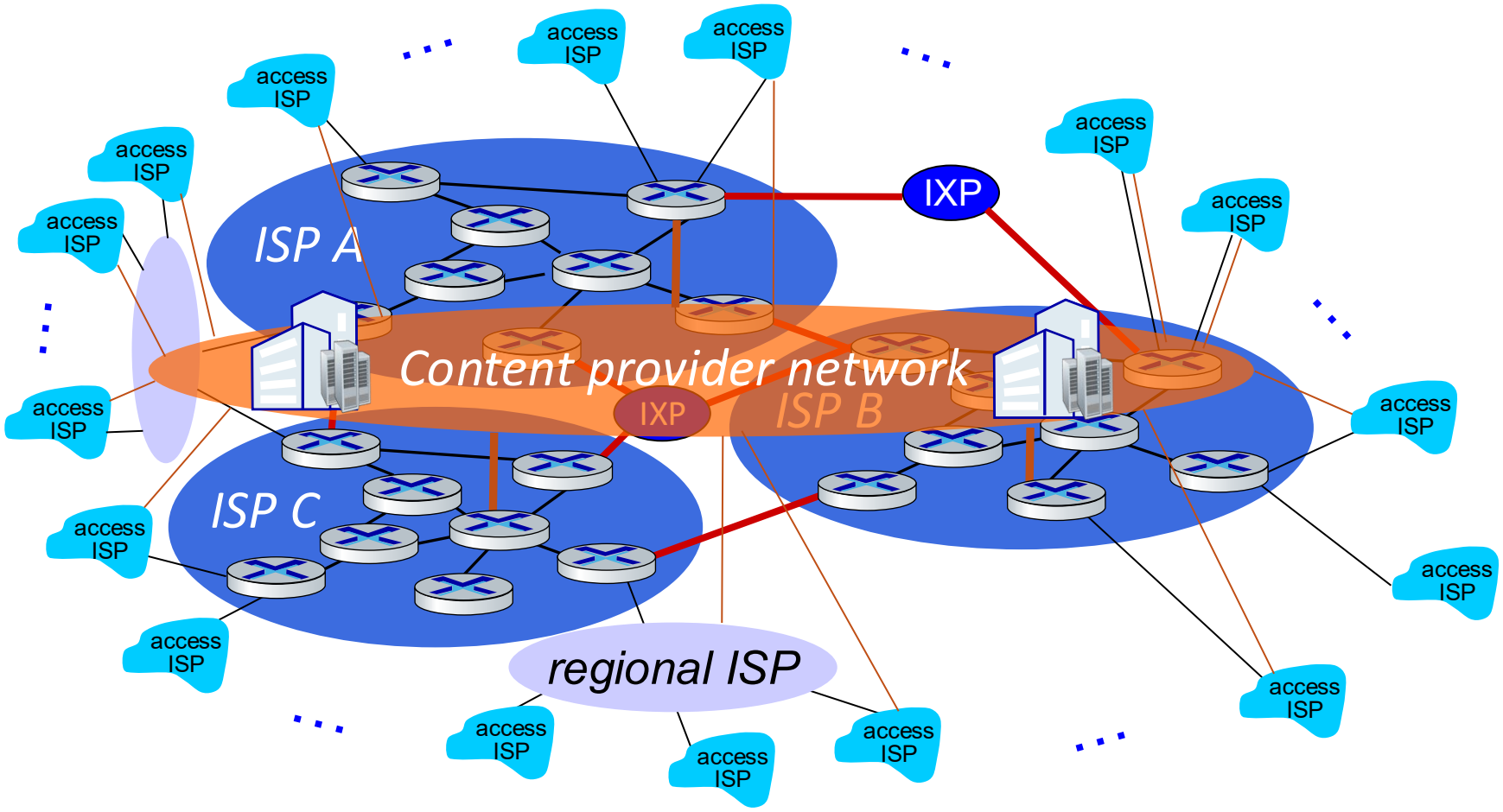
Internet structure: a “network of networks”

… and regional networks may arise to connect access ISPs to larger ISPs

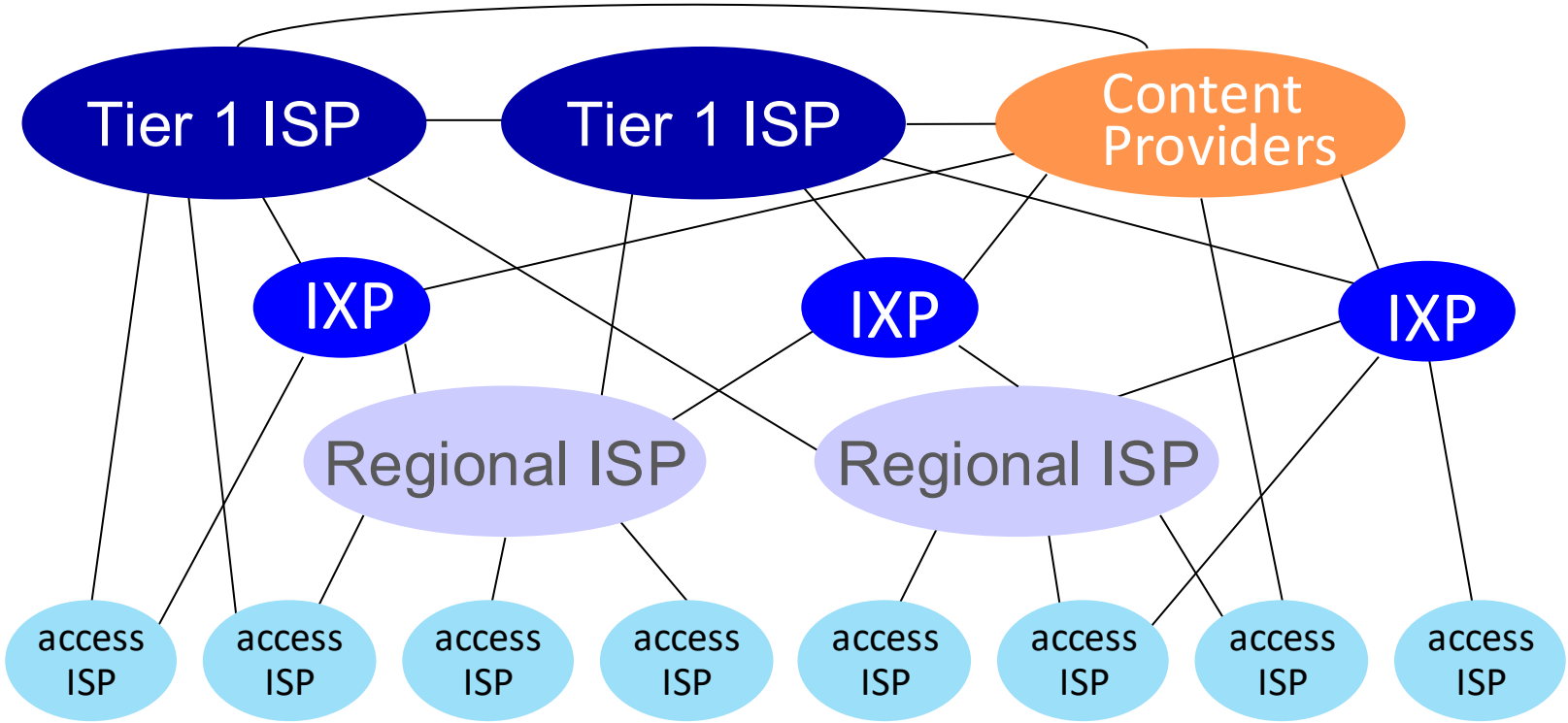


Internet structure: a “network of networks”

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: a "network of networks"



At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Review questions

- Explain what is meant by the following terms
  - Host
  - Communication link
  - ISP
  - IXP
  - Content provider
- What are the devices in a network that can interconnect hosts or networks?
- Find some other network extents (xAN) not covered in class.

Lesson outline

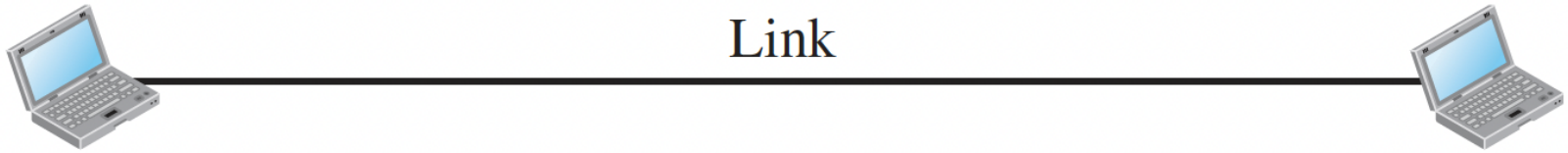
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Network topologies

- Point-to-point
- Bus
- Ring
- Star (hub and spoke)
- Tree
- Mesh

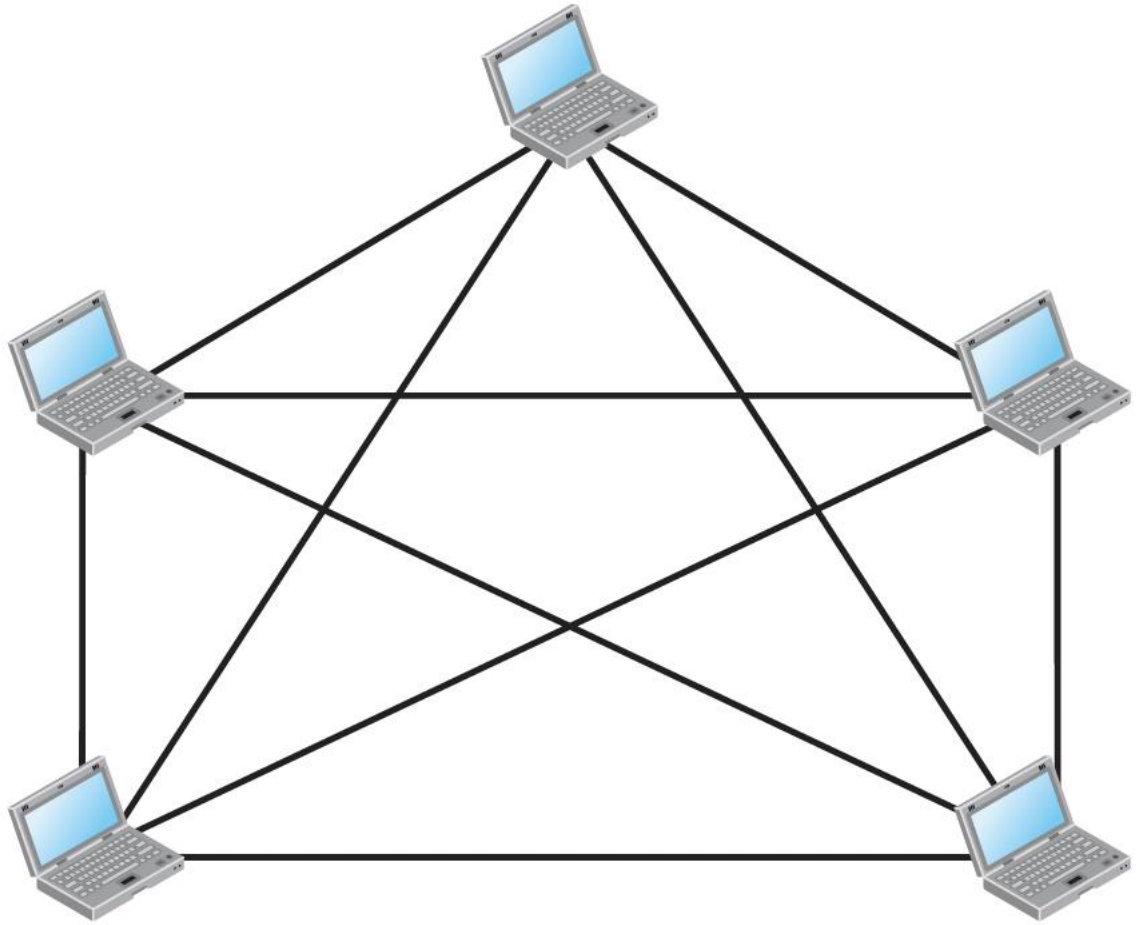
Point-to-point

- A point-to-point connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.



Mesh

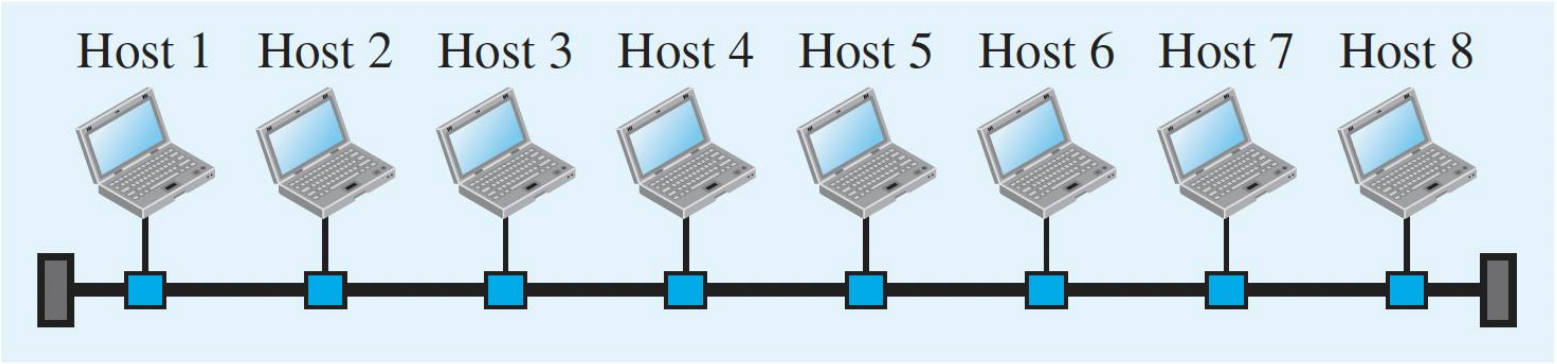
- Devices have dedicated point-to-point links to all or many other devices
- Full-mesh connectivity: every device is connected to every other device
- If the number of nodes is N, full-mesh connectivity requires N(N-1)/2 links
- Usually full-mesh connectivity is avoided as it is expensive in terms of the number of links
- Partial-mesh: Number of links less than full-mesh



No. of links = N(N-1)/2

Bus

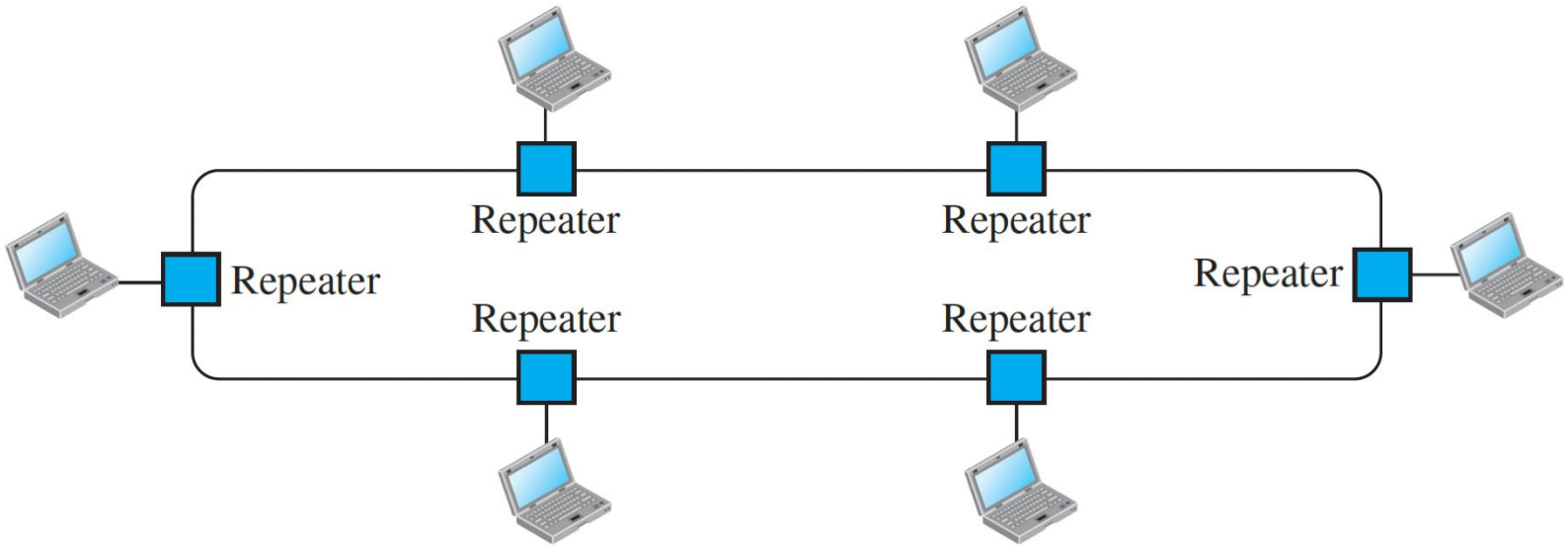
- All stations attach directly to a linear transmission medium, or bus
- Usually through appropriate hardware interfacing known as a tap
- Signal transmitted from any station propagates the full length of the medium in both directions and can be received by all other stations
- At each end of the bus is a terminator, which absorbs any signal, removing it from the bus



No. of links = 1

Ring

- Each device has a dedicated point-to-point connection with only the two devices on either side of it
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination
- Each device in the ring has a repeater, which receives the bits, regenerates them and passes them along

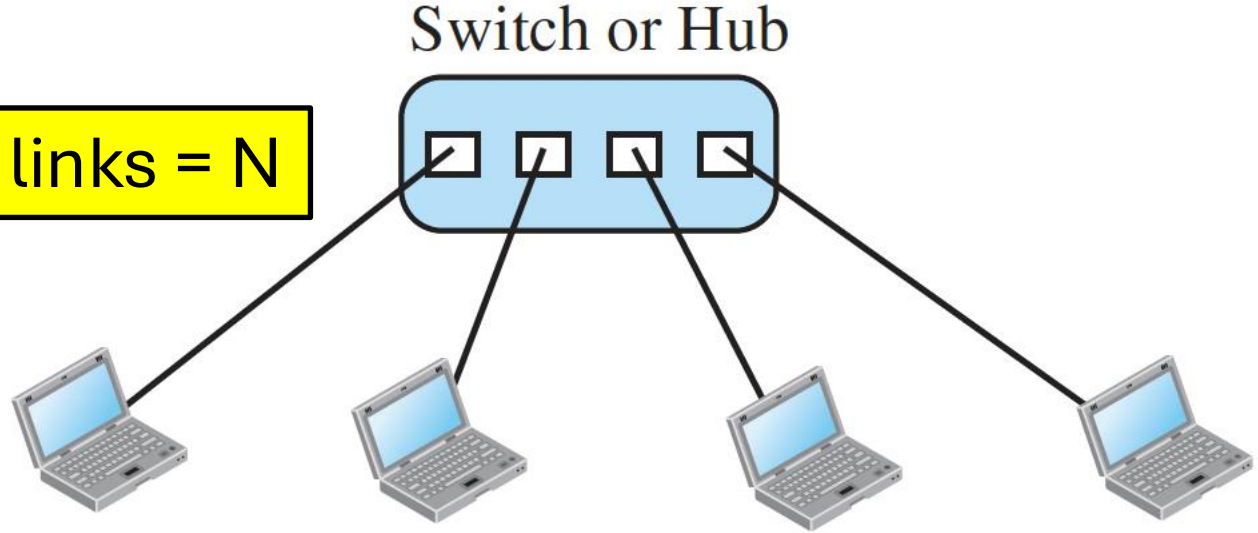


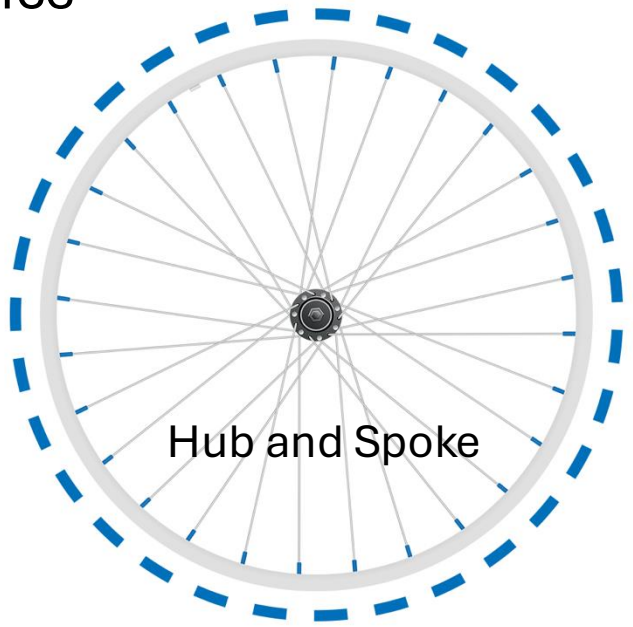
No. of links = N

Star (hub and spoke)

- Each device has a dedicated point-to-point link only to a central controller
- Star topology does not allow direct traffic between devices
  - The devices are not directly linked to one another
  - Everything must go through the controller
- The controller acts as an exchange
  - If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device

No. of links = N

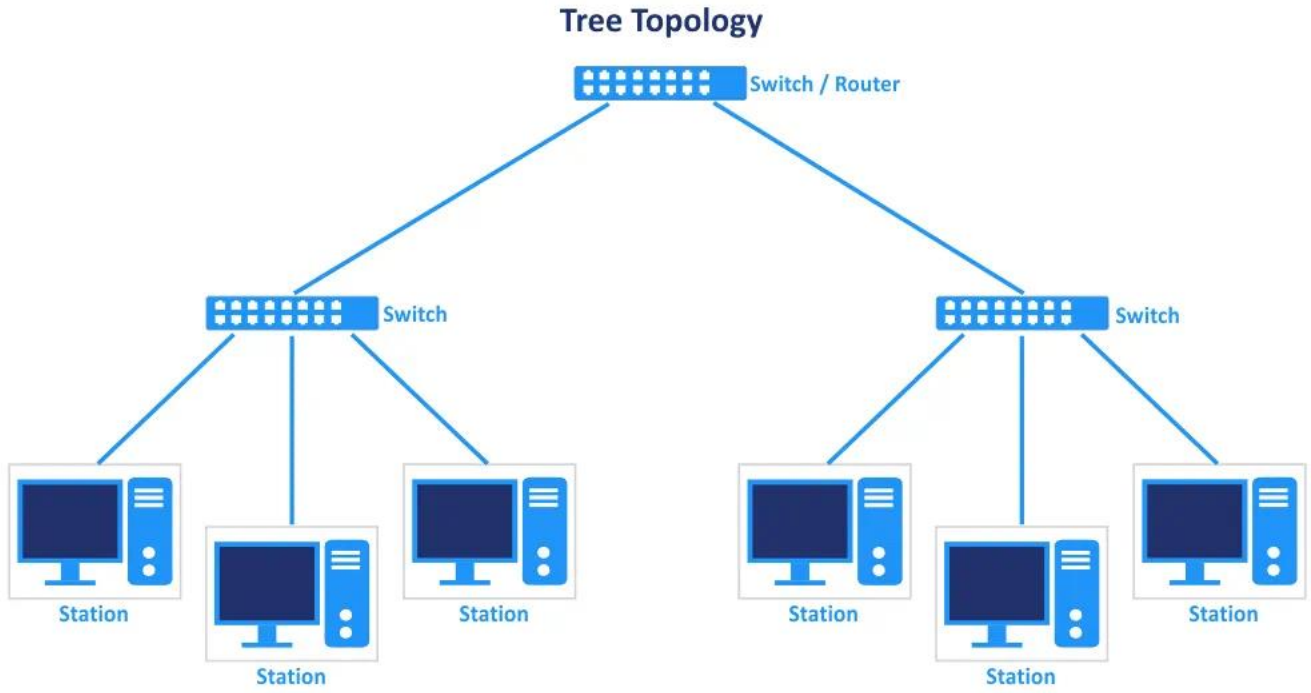




Tree

- Multiple levels of star type networks
- Usually, end stations are at the bottom level
- Upper levels are used for interconnection of switches and routers
  - No end stations connect to upper layer devices

No. of links > N



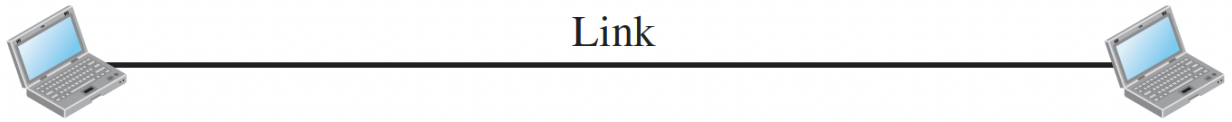
Review questions

- Describe the point-to-point, mesh, bus, ring, star, and tree topologies of networks
- Which of these can have a tree structure
  - LAN
  - Campus network
  - PAN
  - MAN
  - WAN
- Is it possible to have a private communication (that others cannot see/hear) between two devices in a network with bus topology?
- What is the number of links in a full-mesh network of 4 devices
- What is the number of links in a full-mesh network of 100 devices

Lesson outline

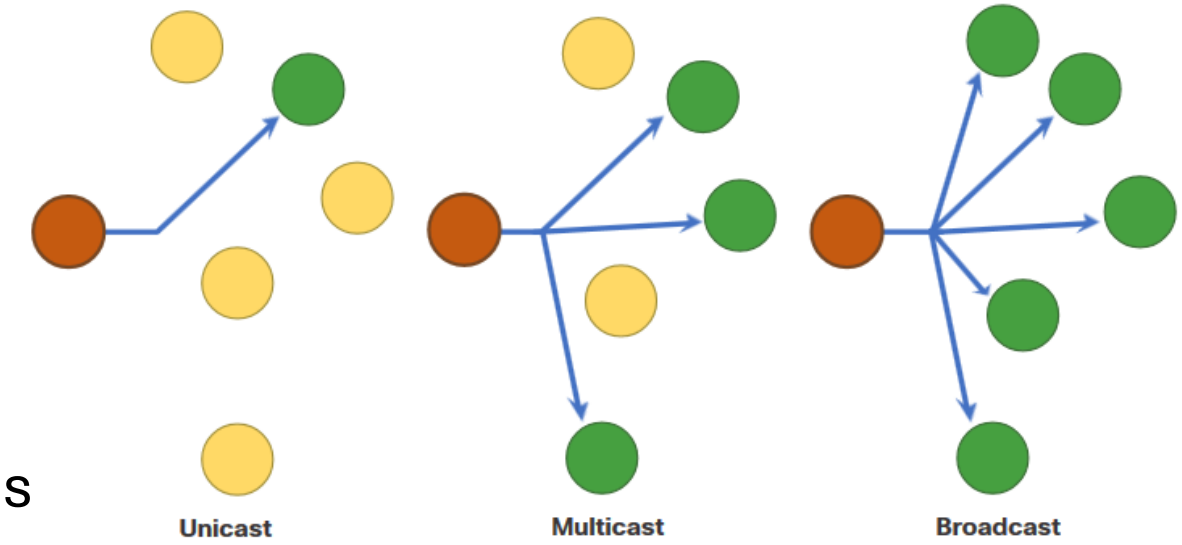
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Data flow in a link (between two devices)



- Simplex
  - Communication is unidirectional
  - Only one of the two devices on the link can transmit; the other can only receive
  - Keyboards and traditional monitors are examples of simplex devices
  - The keyboard can only introduce input; the monitor can only accept output
- Half-Duplex
  - Each station can transmit and receive, but not at the same time
  - When one device is sending, the other can only receive, and vice versa
  - Walkie-talkies are half-duplex systems
- Full-Duplex
  - Both stations can transmit and receive simultaneously
  - Data flows in the two directions share the communication channel (link)
  - Two separate “paths” must be available for data flow in the two directions
  - Example: Telephone network where it is possible for both parties to talk at the same time

Data flow in networks



- Unicast
  - From the sender to one destination
- Broadcast
  - From the sender to all destinations (devices) in the network
  - Useful in making “announcements”
  - Usually limited to the local network to which the sender is connected*
- Multicast
  - From the sender to selected destinations
  - Hosts must join a multicast group to receive data sent to the group
  - Members of the group may be in sender’s network or other networks

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Circuit Switching

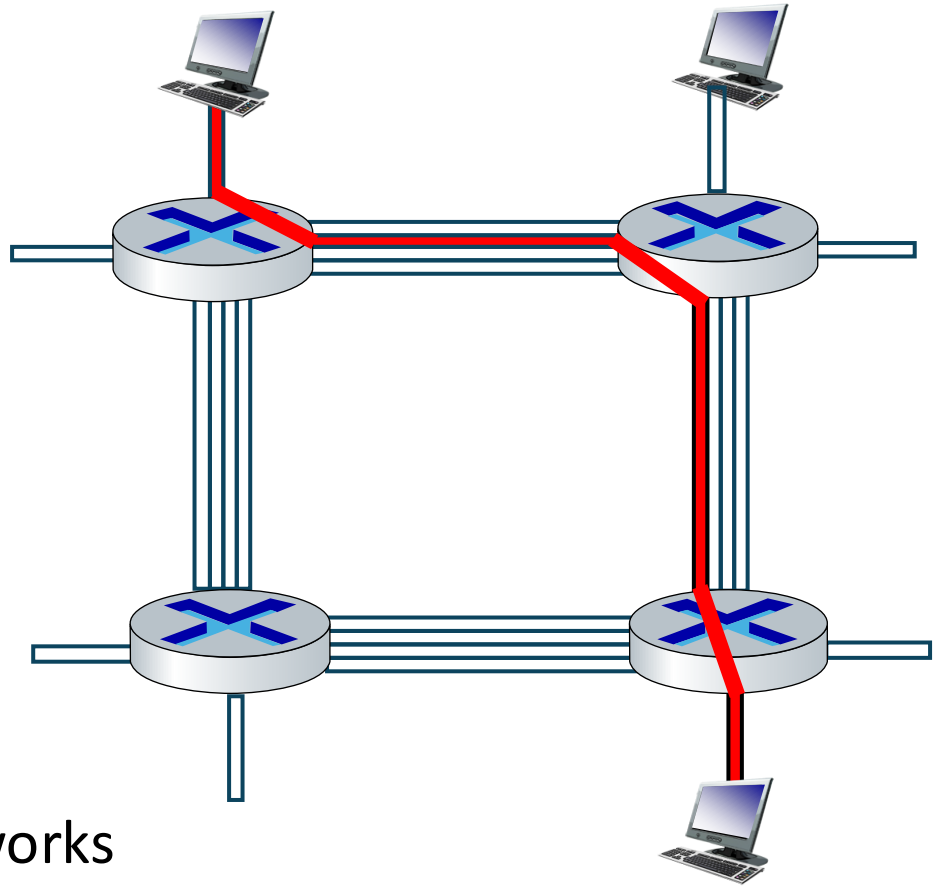
- Uses a dedicated path between two stations
- Has three phases
  - Connection establishment
  - Data transfer
  - Disconnection
- Can be inefficient
  - Channel capacity dedicated for duration of connection
  - If no data, capacity is wasted
- Set up (connection) takes time
- Once connected, transfer is transparent



Circuit switching

end-end resources allocated to, reserved for "call" between source and destination

- in diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call (no sharing)
  - commonly used in traditional telephone networks



* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive

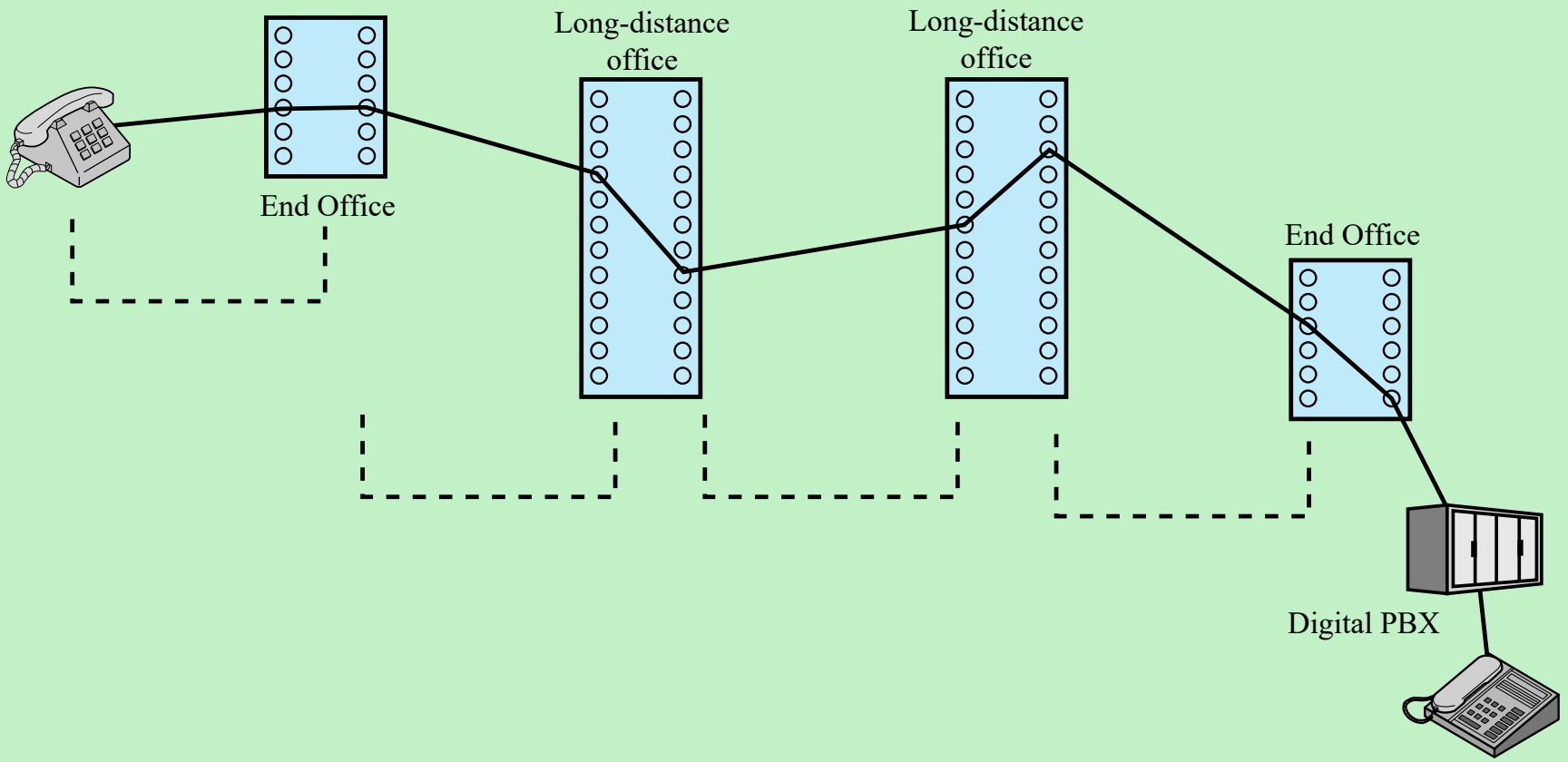
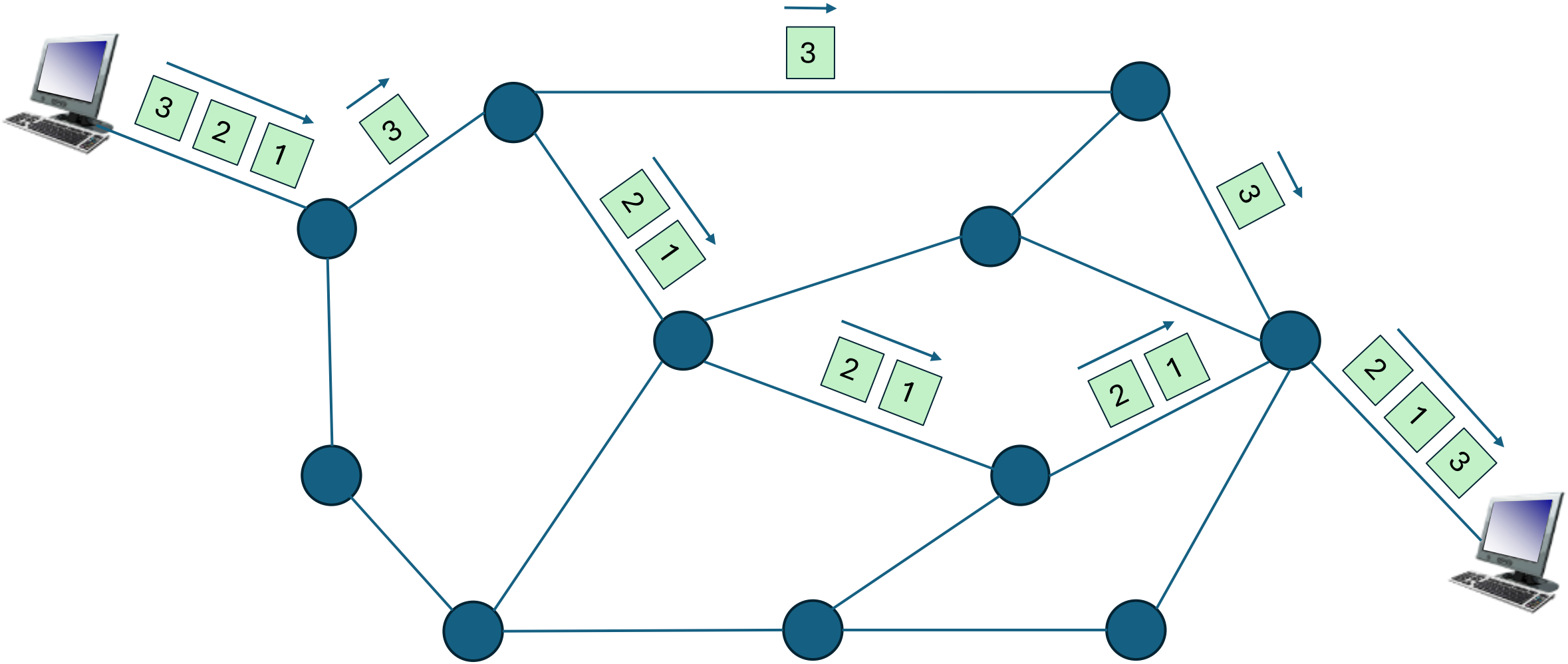


Figure 9.2 Example Connection Over a Public Circuit-Switching Network

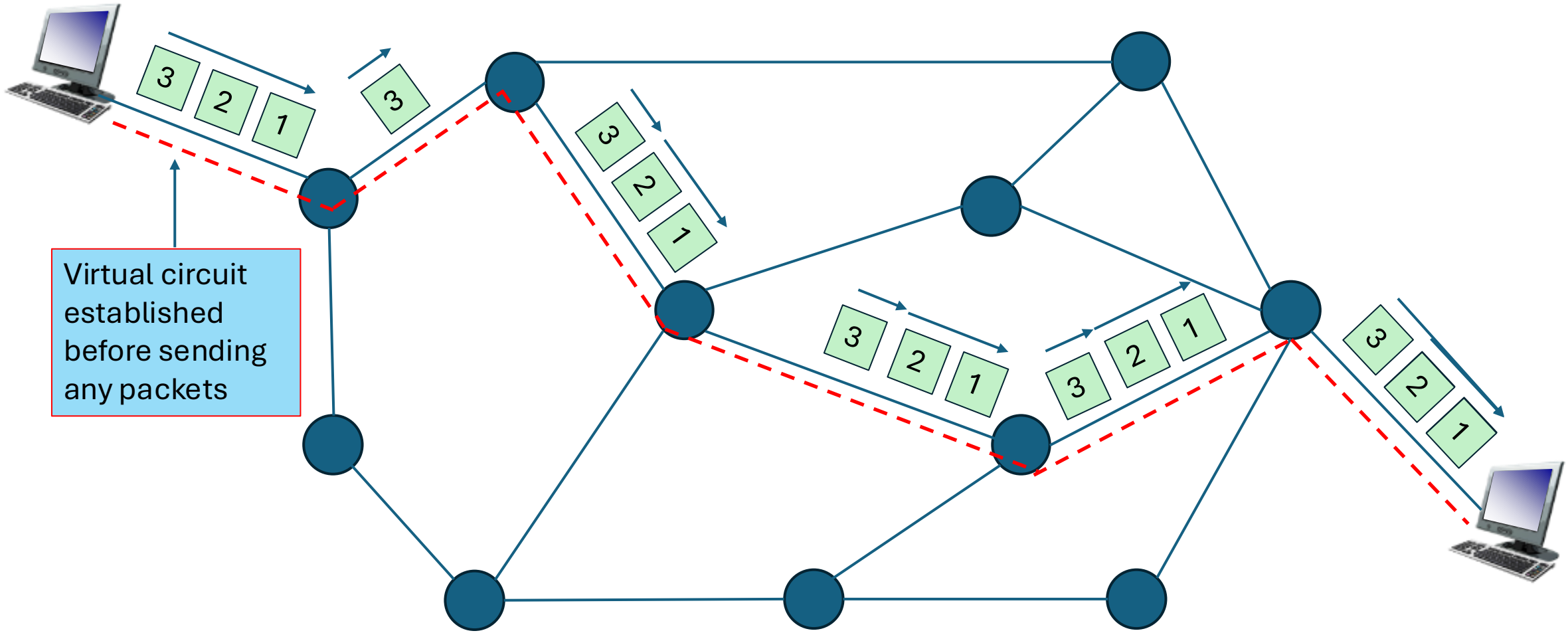
Packet Switching

- Data transmitted in small packets
  - Circuit switching was designed for voice
  - Packet switching was designed for data
- Packets contain user data (payload) and control information (header)
  - Header includes information like sender, recipient addresses
- Packets are received, stored briefly (buffered) and forwarded to the next node
- Datagram approach
  - Each packet is treated independently with no reference to previous packets
  - Packets may be routed via different paths
  - Packets may arrive out of order
- Virtual circuit approach
  - A preplanned route is established before any packets are sent (like circuit switching)
  - All packets routed over the same path
  - Packets will arrive in the order in which they are sent

Packet Switching: Datagram approach



Packet Switching: Virtual circuit approach



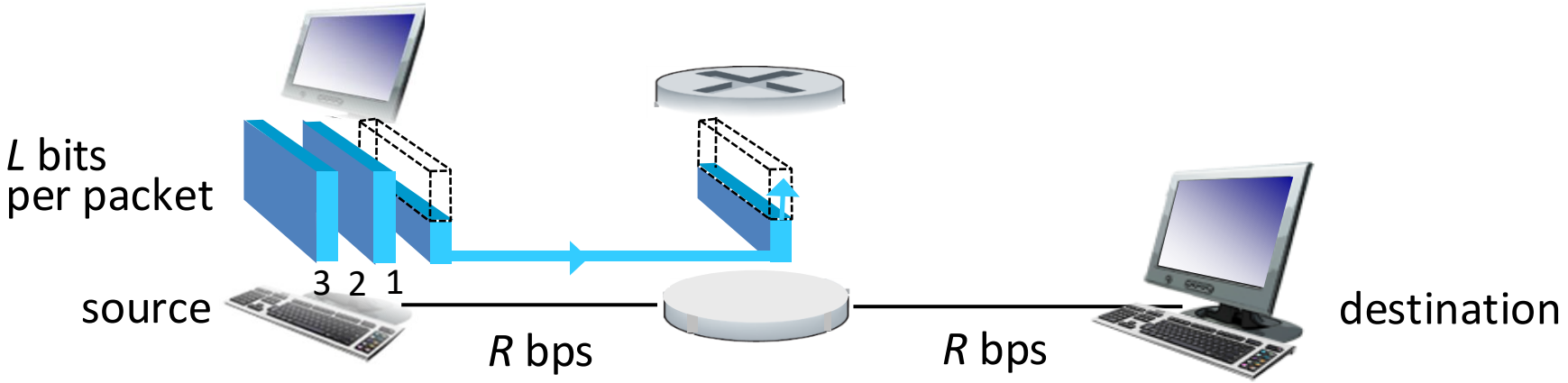
Review questions

- Which type of network data flow (unicast, broadcast, multicast) would be most suitable for the following applications
  - Live streaming of a campus event over the campus network
  - Checking to see a specific device (with known name/address) is online
  - Trying to find a server on the network which can provide a specific service
- Which type of data flow (simplex, half duplex, full duplex) would occur in the following scenarios
  - Television or FM radio broadcasting
  - Communication between two devices in a Wi-Fi network

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Packet-switching: store-and-forward

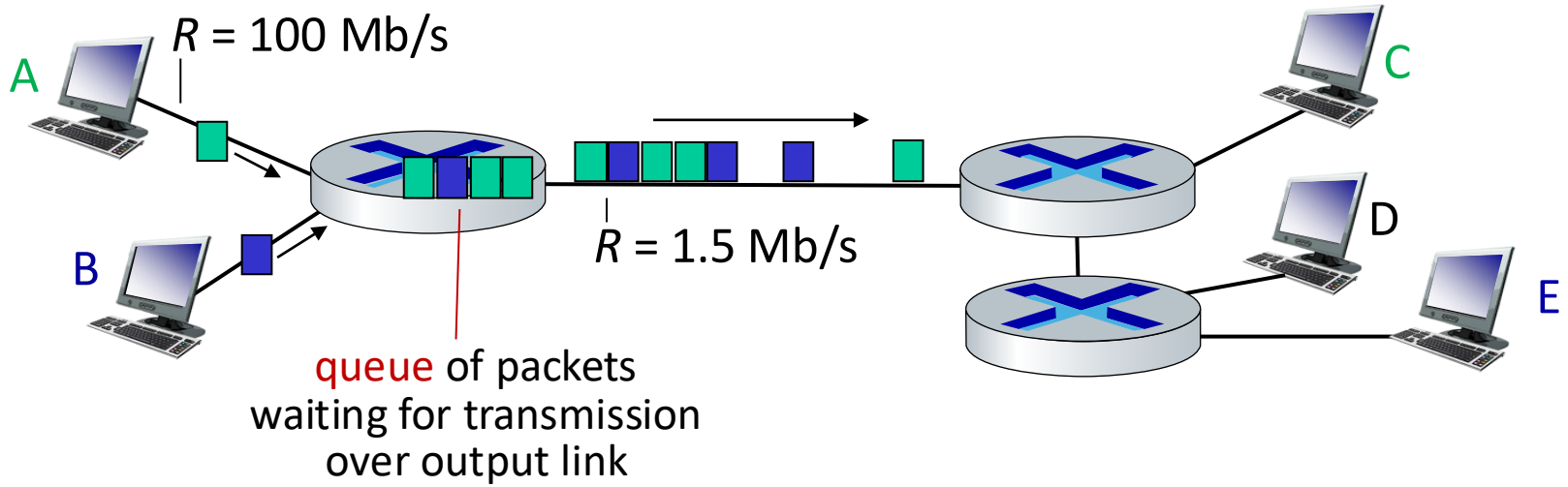


- packet transmission delay: takes L/R seconds to transmit (push out) L-bit packet into link at R bps (bits per second)
- *store and forward: entire* packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- L = 10 Kbits
- R = 100 Mbps
- one-hop transmission delay = 0.1 msec

Packet-switching: queueing



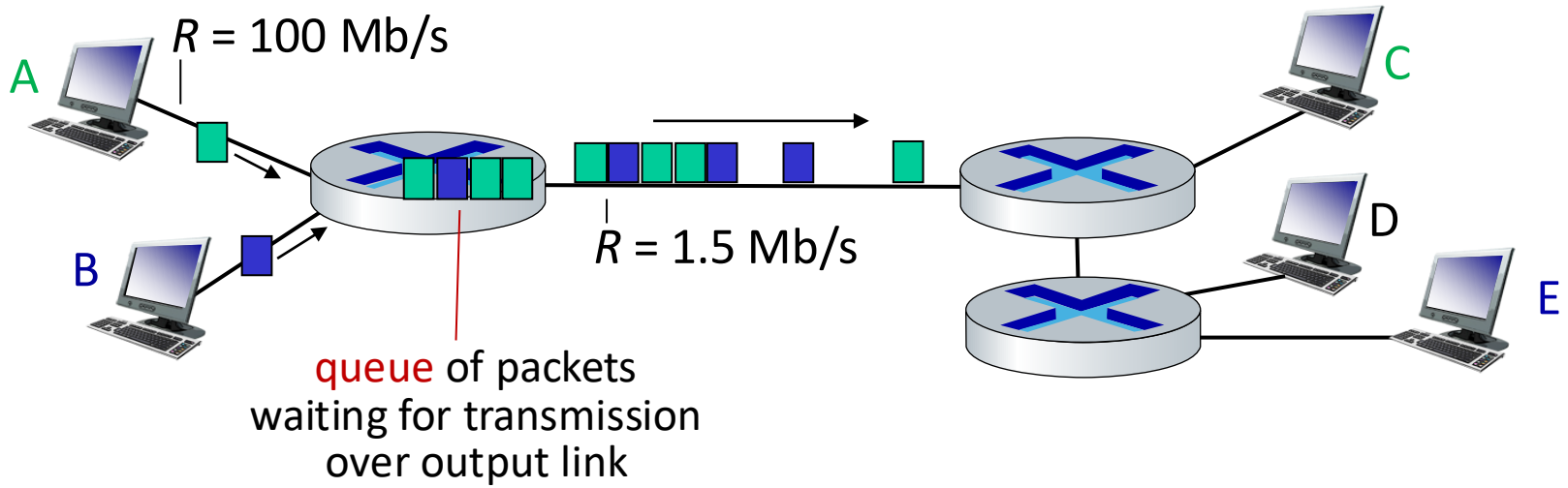
Queueing occurs when work arrives faster than it can be serviced:







Packet-switching: queueing

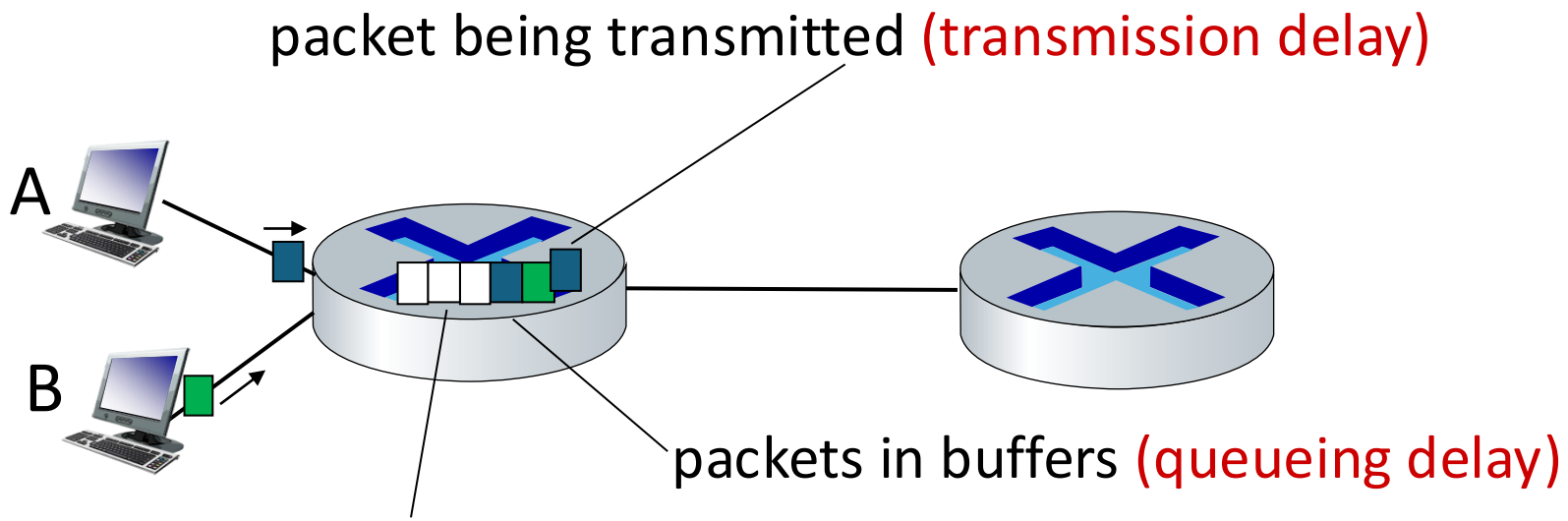


Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

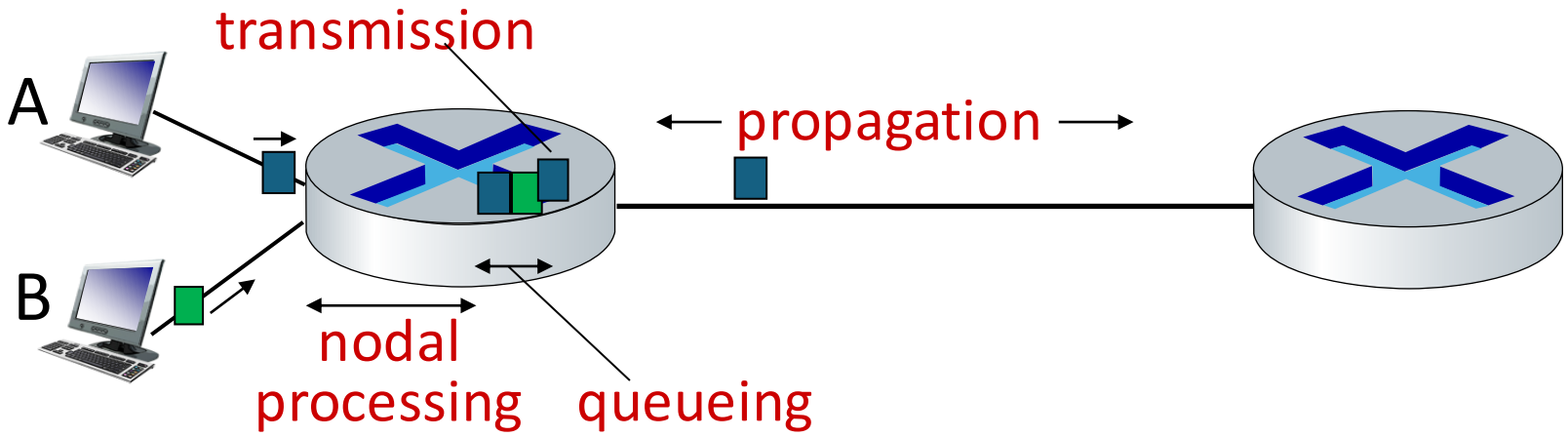
- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

How do packet delay and loss occur?

- packets queue in router buffers, waiting for turn for transmission
  - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
- packet *loss* occurs when memory to hold queued packets fills up



Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

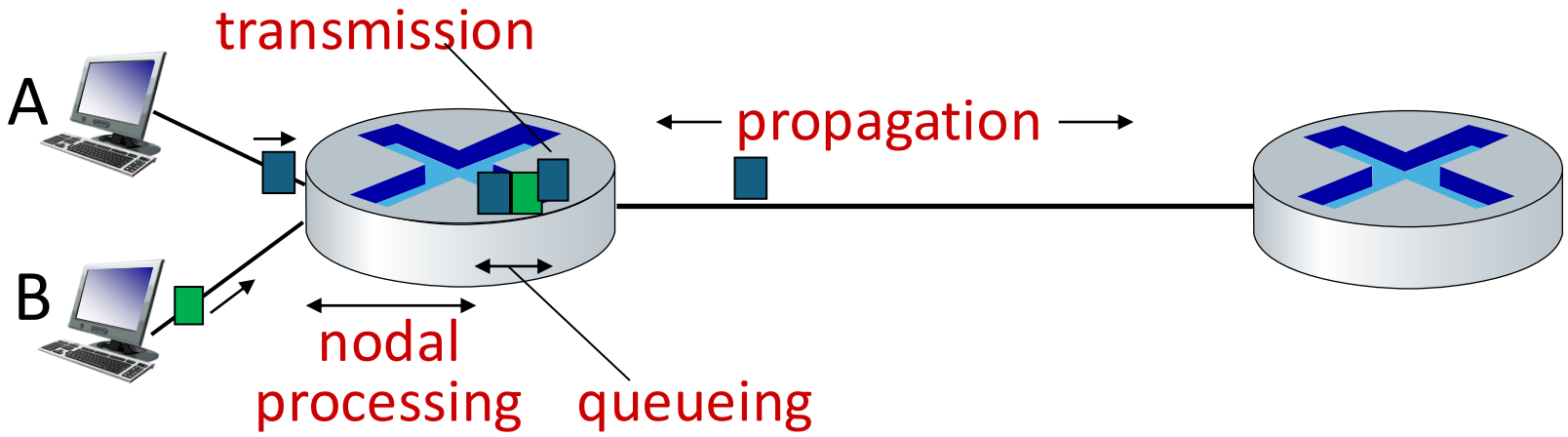
d_{proc}: nodal processing

- check bit errors
- determine output link
- typically microseconds

d_{queue}: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet delay: four sources



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

d_{trans}: transmission delay:

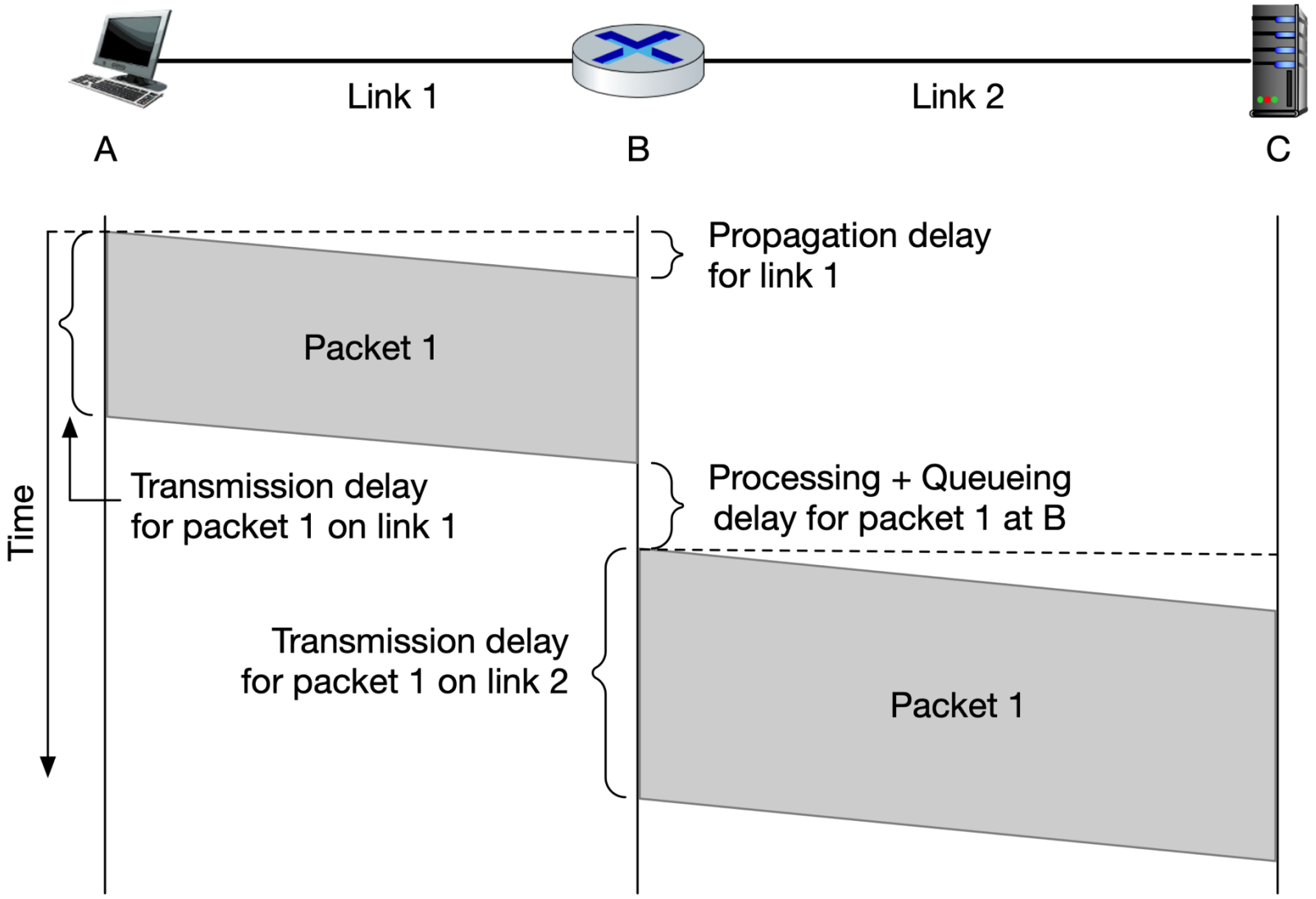
- L: packet length (bits)
- R: link *transmission rate (bps)*
- $d_{trans} = L/R$

d_{prop}: propagation delay:

- d: length of physical link
- s: propagation speed (~2×10^8 m/sec)
- $d_{prop} = d/s$

d_{trans} and d_{prop} *very* different

Timing diagram showing packet flow

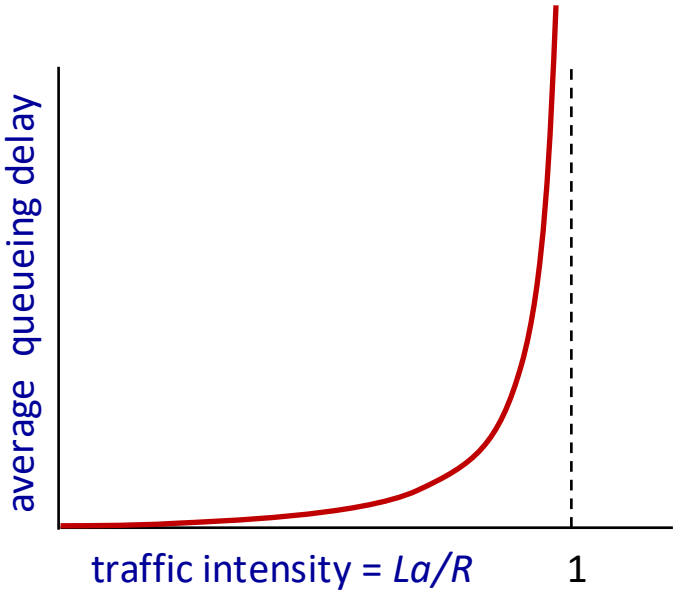


Packet queueing delay (revisited)

- *a:* average packet arrival rate
- *L:* packet length (bits)
- *R:* link bandwidth (bit transmission rate)

$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}}$$

"traffic intensity"



- *La/R* ~ 0: avg. queueing delay small
- *La/R* → 1: avg. queueing delay large
- *La/R* > 1: more "work" arriving is more than can be serviced - average delay infinite!



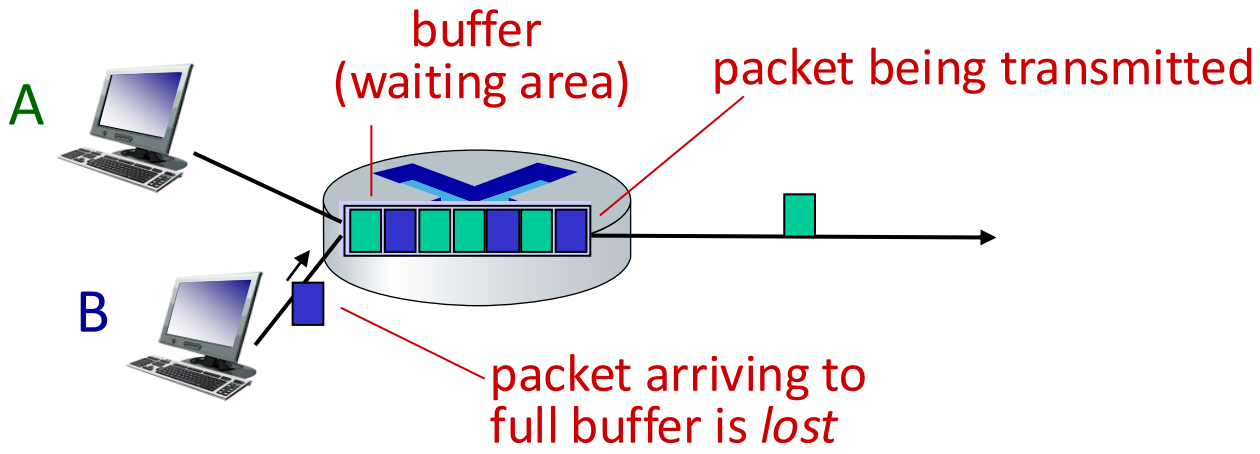
La/R ~ 0



La/R -> 1

Packet loss

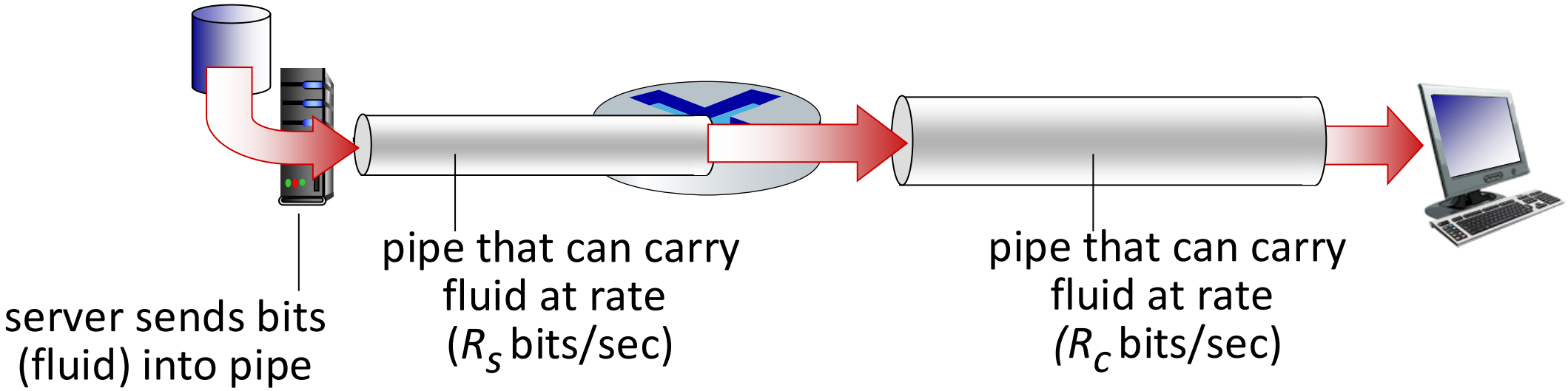
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



* Check out the Java applet for an interactive animation (on publisher's website) of queuing and loss

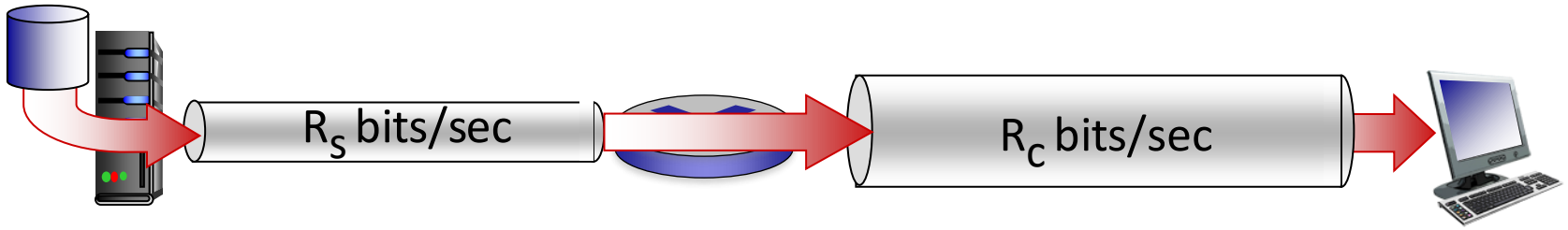
Throughput

- *throughput:* rate (bits/time unit) at which bits are being sent from sender to receiver
  - *instantaneous:* rate at given point in time
  - *average:* rate over longer period of time

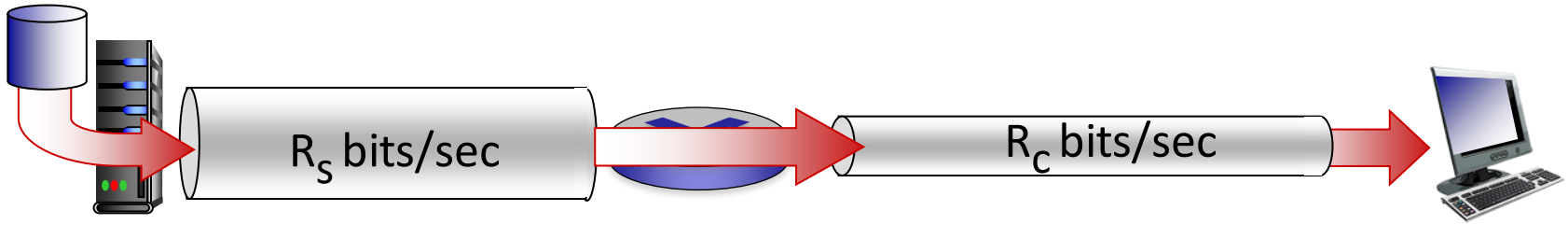


Throughput

$R_s < R_c$ What is average end-end throughput?



$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

Review questions

- A packet of length 1,000 bytes is sent over a link of distance 2,500 km and transmission rate 2 Mbps. The propagation speed of the medium is 2.5×10^8 m/s.
  - How long does it take for the packet to reach the other end?
  - Does this delay depend on packet length?
  - Does this delay depend on transmission rate?
- Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links, of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps.
  - Assuming no other traffic in the network, what is the throughput for the file transfer?
  - How long will it take to transfer a file of 4 million bytes to Host B?

Lesson summary

- Networks
- Network extent (sizes)
- Network topologies
- Data flow in networks
- Circuit switching and packet switching
- Performance: Loss, Delay, Throughput

References

- Chapter 1 - Introduction
  James F. Kurose and Keith W. Ross, Computer Networking – A Top-Down Approach, (8th Edition), Pearson, 2020
- Chapter 1 - Data Communications, Data Networks, and the Internet
  William Stallings, Data and Computer Communications (10th Edition), Pearson, 2014

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